Date 21 | 02 | 2016 23

Sequences

Arithmetic progression (AP)

n^{th} term $\longrightarrow$ $T_n = a + (n-1)d$

Sum of first n terms $\longrightarrow$ $S_n = \frac{1}{2} n [2a + (n-1)d]$

$\longrightarrow$ $S_n = \frac{1}{2} n (a + l)$

a = first term , l = last term , n = number of terms
d = common difference

(01) consider the sequence 2, 4, 6, 8, ...

a) Find the 20th term :-

* $T_{20} = 2 + (20-1)2 = \underline{\underline{40}}$

b) Find the sum of first 15 terms S_{15} :-

* $S_{15} = \frac{1}{2} \times 15 [2 \times 2 + (15-1)2]$

$= \frac{15 \times 32}{2}$

$= \underline{\underline{240}}$

02. It is planned to arrange chairs in an outdoor event in a triangular shaped area. first row should have 1 chair and every row should have 2 more chairs than the previous. If there are 35 chairs, how many rows will be there.

$S_n = \frac{1}{2} n(2a + (n-1)d)$

$35 = \frac{1}{2} n(2 \times 1 + (n-1)2)$

$70 = 2n + 2n(n-1)$

$70 = 2n + 2n^2 - 2n$

$0 = 2n^2 - n - 70$

$n = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 2 \times (-70)}}{2 \times 2}$

$n = \frac{1 \pm \sqrt{1 + 840}}{4}$

$n = \frac{1 \pm 29}{6}$

$n = \frac{1+29}{6}$ or $n = \frac{1-29}{6}$

$n = 5$ $n = \frac{-28}{6}$

$n \neq \frac{-28}{6}$

$\therefore n = 5$

* there are 5 rows

03. consider the number pattern 3, 5, 7, 9 . . .

a) Find the 25th value in this pattern.

$T_{25} = 3 + (25-1)2$

$= 3 + 48$

$= 51$

b) Find the sum of ~~#~~ first 15 values.

* $S_{15} = \frac{1}{2} 15(2 \times 3 + (15-1)2)$

$= 15 \times 17 \quad = 255$

Geometric Progression (GP)

n^{th} Term $\longrightarrow$ $T_n = ar^{(n-1)}$

Sum of first n terms $\longrightarrow$ $S_n = \frac{a(1-r^n)}{(1-r)}$

if $(-1) < r < 1$ then $\longrightarrow$ $S_\infty = \frac{a}{(1-r)}$

$(1/2)^{1000\ldots} \doteq 0$

$(1/2)^{\infty} \doteq = 0$

r = common ratio

01. consider the number sequence 3, 6, 12, 24 . . .

a) find the 12th term :

* $T_{12} = 3 \times (2)^{12-1} \quad = 3 \times 2^{11} \quad = 3 \times 2048 = 6136$

b) find the sum of the first 10 terms :

* $S_{10} = \frac{3(1-2^{10})}{(1-2)}$

$= \frac{3(-1023)}{(-1)} \quad = 3069$

4096
2
2

LMS

01. $T_n = \frac{1}{2}a + (n-1)d$

$T_{13} = 2 + (13-1)2 = \underline{26}$

8191
× 2
16382

02. $T_{10} = 1 + (10-1)2 = 19$

03. $S_{10} = \frac{1}{2} \times 10\,(2\times 1 + (10-1)2)$

$= 10\,(10) = 100$

04. $T_{13} = 2 \cdot 2^{(13-1)} = 2 \times 2^{12} = 8192$

05. $S_{13} = \frac{2(1-2^{13})}{1-2} = 2 \times 8191 = 16382$

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01. consider the number pattern 2, 2, 4, 4, 8, 6, 16, 8 ...

a) Find the 21st value :-

* $T_{21} = 2 \times 2^{(21-1)} = 2^{\frac{1}{21}} = \underline{2048}$

b) Find the sum of first 15 values :-

* $S_{15} = \frac{1}{2}(7(2\times 2 + (7-1)2)) + \frac{2(1-2^8)}{(1-2)}$

$= 56 + 510$

$= \underline{566}$

Recursive sequence

* sequence is given by,

$$T_n = T_{n-1} + T_{n-2}$$

where $T_1 = 1$ and $T_2 = 1$ write the first 5 values of this sequence.

* $T_5 = T_4 + T_3$

$T_4 = T_3 + T_2$

$T_3 = T_2 + T_1$

$T_3 = 1 + 1 = 2$

$T_4 = 2 + 1 = 3$

$T_5 = 3 + 2 = 5$

ex: $f(x) = f(x-2) + 2$ where $f(1) = 3$:
calculate $f(5)$;

* $f(5) = f(4) + 2$
$f(4) = f(3) + 2$
$f(3) = f(2) + 2$
$f(2) = f(1) + 2 \quad : 3 + 2 = 5$

$f(3) = 5 + 2 = 7$
$f(4) = 7 + 2 = 9$

$f(5) = 9 + 2 = 11$

28 02 2023

Number conversions

Decimal - Binary conversions

Convert 25 into binary

2 | 25
2 | 12 — 1
2 | 6 — 0
2 | 3 — 0
1 — 1

Like This

11001_2

convert 11001_2 to Decimal

11001_2

$2^4 \quad 2^3 \quad 2^0$

$= 2^4 + 2^3 + 2^0$

$= 16 + 8 + 1$

$= 25$

convert 35 into binary

2 | 35
2 | 17 — 1
2 | 8 — 1
2 | 4 — 0
2 | 2 — 0
1 — 0

$= 100011_2$

L.M.S

100011

convert 100011_2 back to decima

100011_2

$2^5 \quad 2^1 \quad 2^0$

$= 2^5 + 2^1 + 2^0$

$= 32 + 2 + 1$

$= 35$

convert 11001_2 back to decimal

11001_2

$2^4 \; 2^3 \; 2^0$

2^{4+3+0}

$= 2^4 + 2^3 + 2^0$

$= 16 + 8 + 1$

$= 25$

Decimal - Octal

convert 123 into octal

8 | 123

8 | 15 — 3

1 — 7

$= 173_8$

convert 25 into octal

8 | 25

3 — 1

$= 31_8$

convert 31_8 into decimal

$3 \; 1 \; 8$

$8^1 \; 8^0$

$= 3 \times 8^1 + 1 \times 8^0$

$= 24 + 1$

$= 25$

convert Binary to Octal

* create groups of 3 starting from right end. Replace each group with one octal digit

method = I

11001_2

$011 \quad 001_2$

$011 \quad 3 \quad 1 \quad +0001_2$

011

2^0

2^1

$2^1 + 2^0$

$2 + 1$

$= 3$

$2^0 = 1$

31_8

method = II

$1\,1\,0\,0\,1_2$

$2^4 \; 2^3 \; 2^2 \; 2^1 \; 2^0$

$16 + 8 + 1$

$= 25$

$8 \mid 25$

$3 \quad 1$

$= 31_8$

Hexadecimal Numbers

number : 0 1 2 3 4 5 6 7 8 9 A B C D E F

Decimal - Hexadecimal

convert 120 to hexadecimal

* $16 \mid 120$

$7 \quad 8$

$= 78_{16}$

L.M.S

78

convert 78_{16} back to decimal

$7^{16^1} \quad 8^{16^0}$

$= 7 \times 16^1 + 8 \times 16^0$

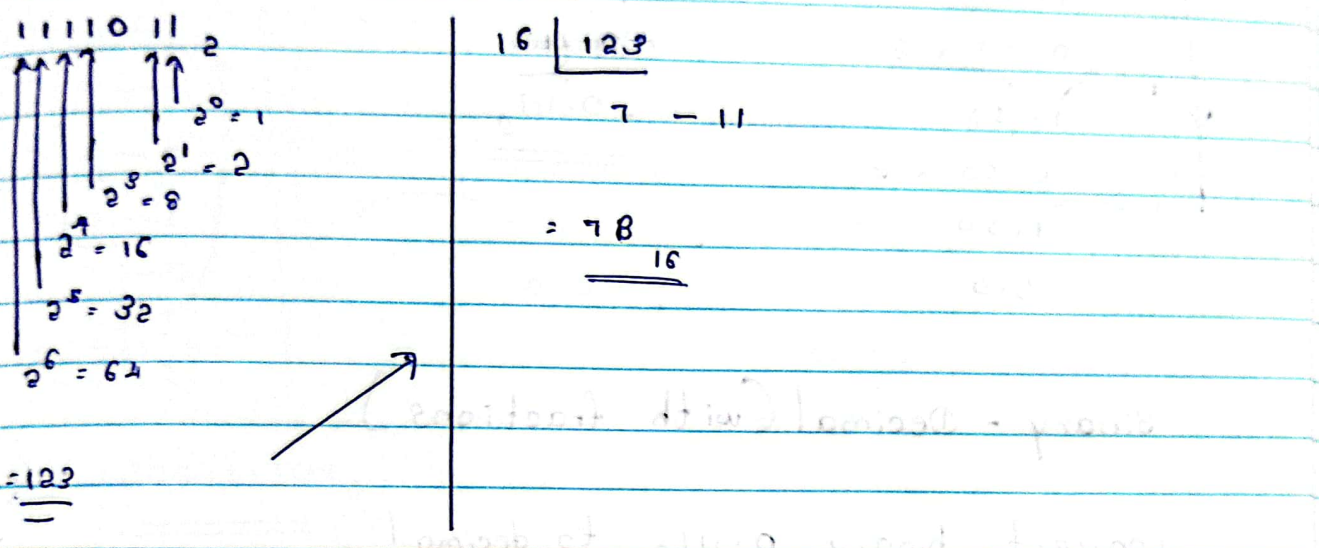
$= 112 + 8$

$= 120$

converting Binary to Hexadecimal

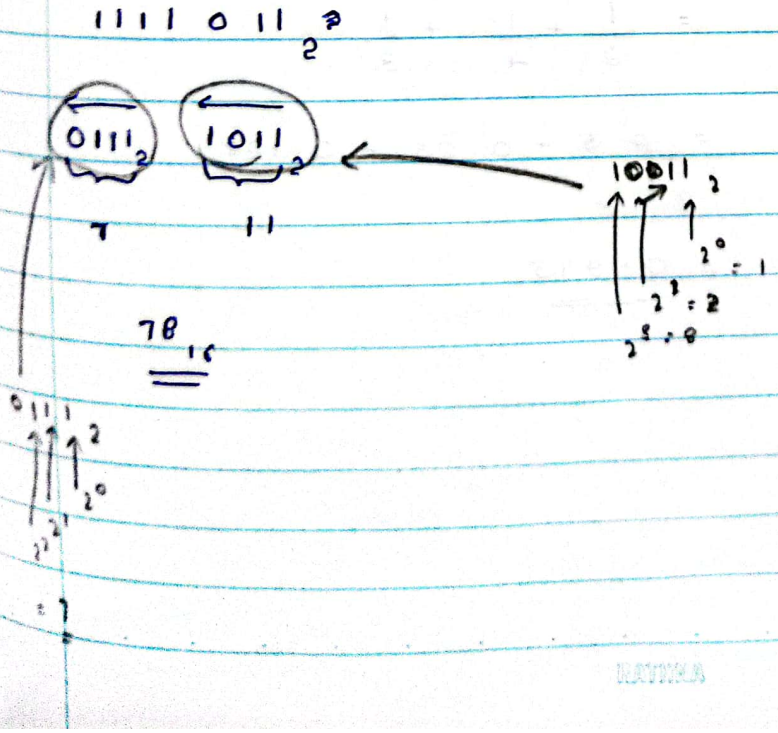
convert 1111011_2 in to hexadecimal

method - I



method - II

* create groups of 4 starting from right end.
* Replace each group with one hexadecimal digit



Decimal - Binary (with fractions)

* convert decimal 0.875 to binary

0.875×2

1 ← 1.750

0.75×2

1 ← 1.50

0.50×2

1 ← 1.00

0.0

$= 0.111_2$

Binary - Decimal (with fractions)

convert binary 0.111_2 to decimal

0.111

2^{-1} 2^{-2} 2^{-3}

$\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

$= 0.5 + 0.25 + 0.125$

$= 0.875$

convert 35.625 to Binary

35.625

2 | 35		0.625 × 2
2 | 17 — 1	1	1.250
2 | 8 — 1		0.25 × 2
2 | 4 — 0	0	0.5 × 2
2 | 2 — 0	1	1.0
1 — 0		0.0

$= 100011_2$ $= 0.101_2$

$\therefore 100011.101_2$

Convert Binary - Octal (with factions)

convert 100011.101_2 to octal

100011.101_2

100 , 011 . 101

100 011 101

2^2 2^1 2^0 2^2 2^0

4 2+1 4+1

43.5_8

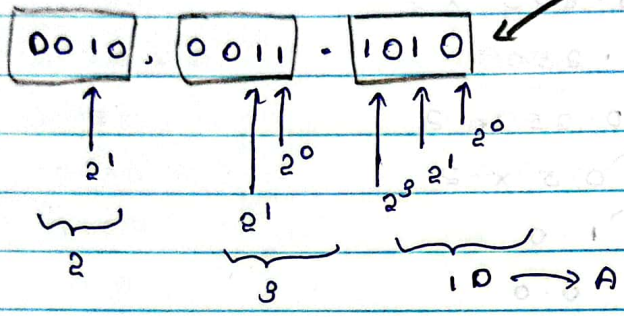
Converting Binary - Hexadecimal (with fractions)

convert 100011.101_2 to hexadecimal

starting from

$1000\ 11 \cdot 101_2$

* Create grps of 4



$= 23.A_{16}$

07.03.2023

convert 37.9375 to

a) Binary ;

* 37.9375

2 | 37		0.9375 × 2
2 | 18 — 1	1	1.8750
2 | 9 — 0		0.8750 × 2
2 | 4 — 1	1	1.7500
2 | 2 — 0		0.75 × 2
1 — 0	1	1.5
		0.5 × 2
	1	1.0

$= 100101_2$ $= 0.1111_2$

ans: 100101.1111

b) octal ;

* $37.9375 = 100101.1111_2$

100 101 . 1111
2

100 101 . 111 100

2^2 2^2 2^0 2^1 2^0 2^2

2^2 4

4 5 7 4

ans = 45.74_8

c) Hexadecimal ;

* $37.9375 = 100101.1111_2$

00100101 . 1111

2^1 2^2 2^0 2^3 2^2 2^1 2^0

2 5 15

ans = $25.F_{16}$

To LMS :-

100101 . 1111

1	0	0	1	0	1	.	1	1	1	1

4	5	.	7	4

25	.	F

01. ~~It is given~~ $x = 257.5793$

a. Find $100x = 25757.93$

b. Find $\frac{x}{1000}$; $\frac{x}{1000} = 0.2575793$

02. It is given that $x = 111011.10100\,1_2$ write $4x$ and $x/8$ in binary without doing any calculations.

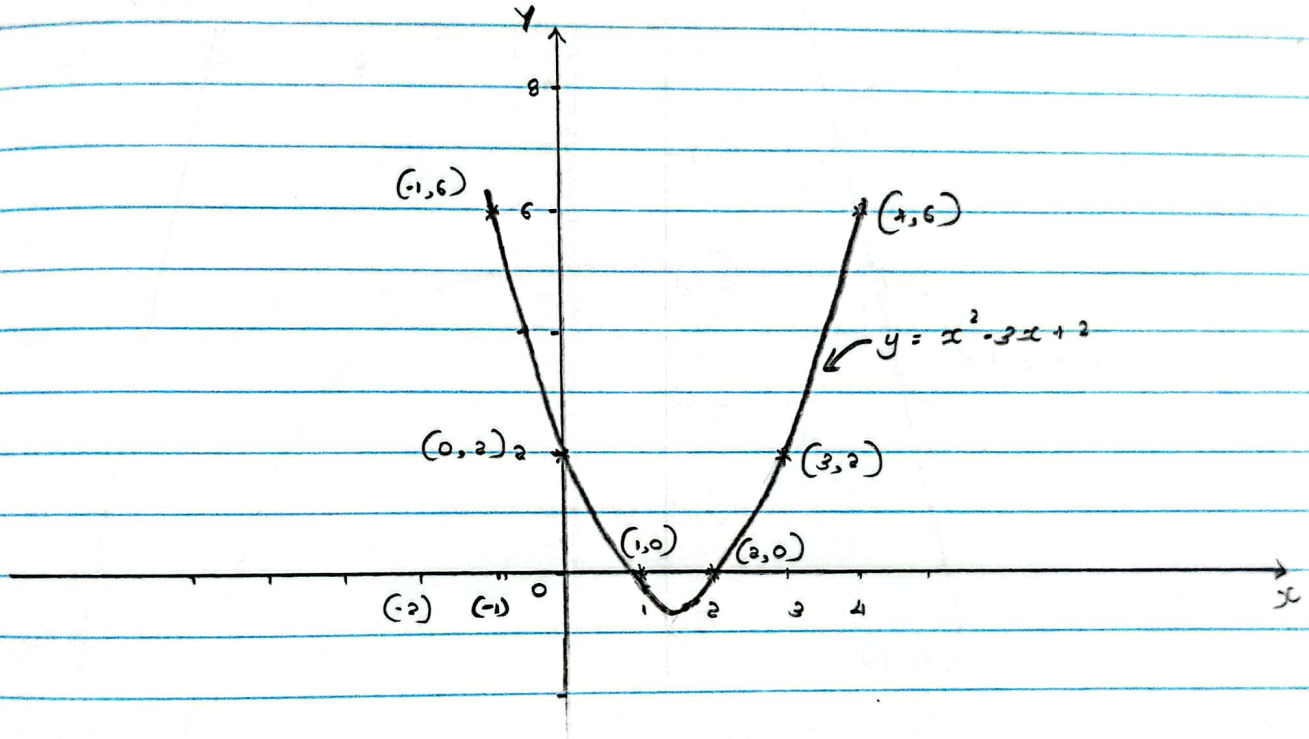
* $4x = 11101110.1001_2$

* $8x/8 = 111.011101001_2$

Drawing Graphs

01. Draw the graph of $y = x^2 - 3x + 2$

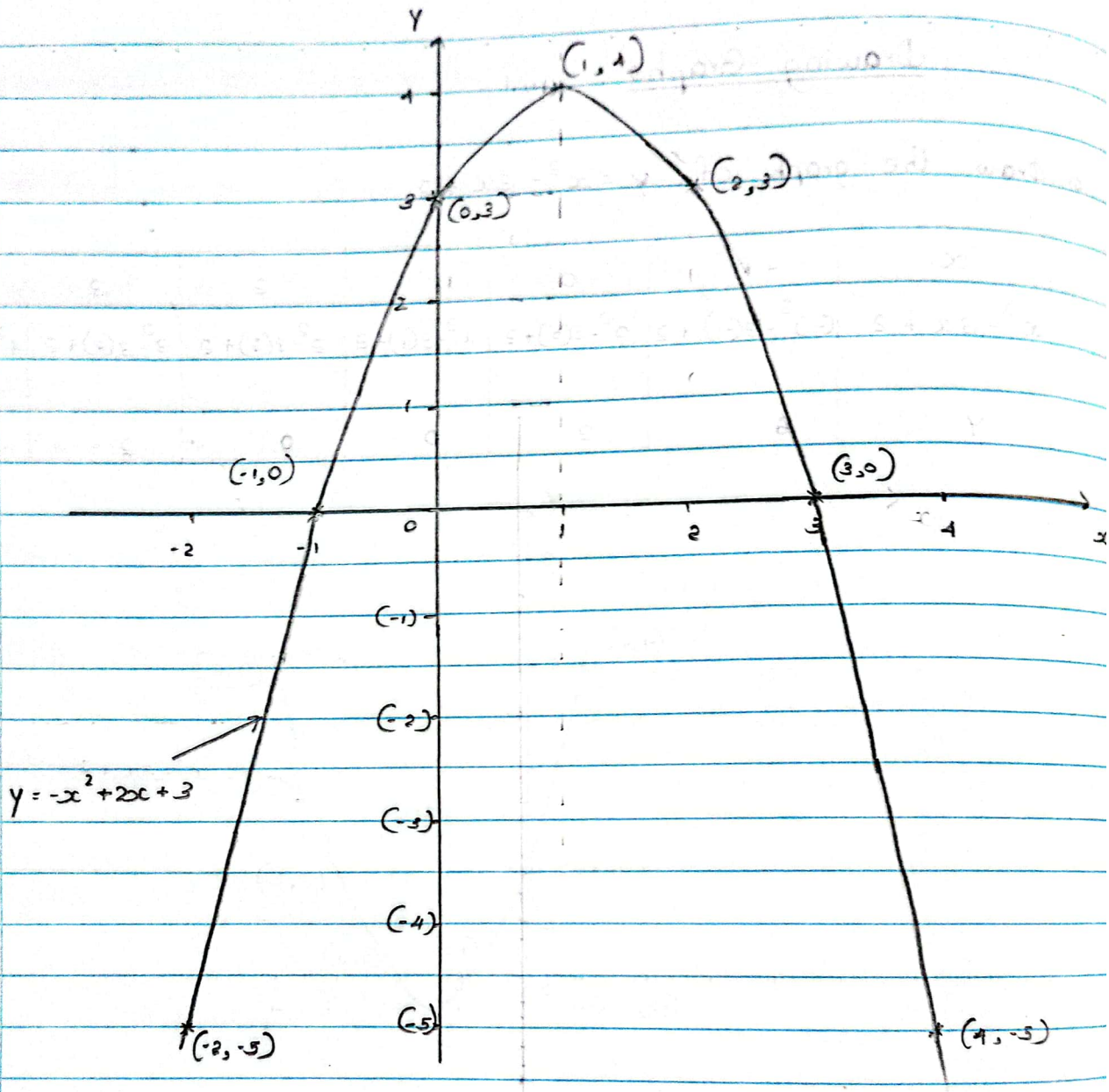
x	-1	0	1	2	3	4
$x^2 - 3x + 2$	$(-1)^2 - 3(-1) + 2$	$0^2 - 3(0) + 2$	$1^2 - 3(1) + 2$	$2^2 - 3(2) + 2$	$3^2 - 3(3) + 2$	$4^2 - 3(4) + 2$
Y	6	2	0	0	2	6



02. $y = -x^2 + 2x + 3$:-

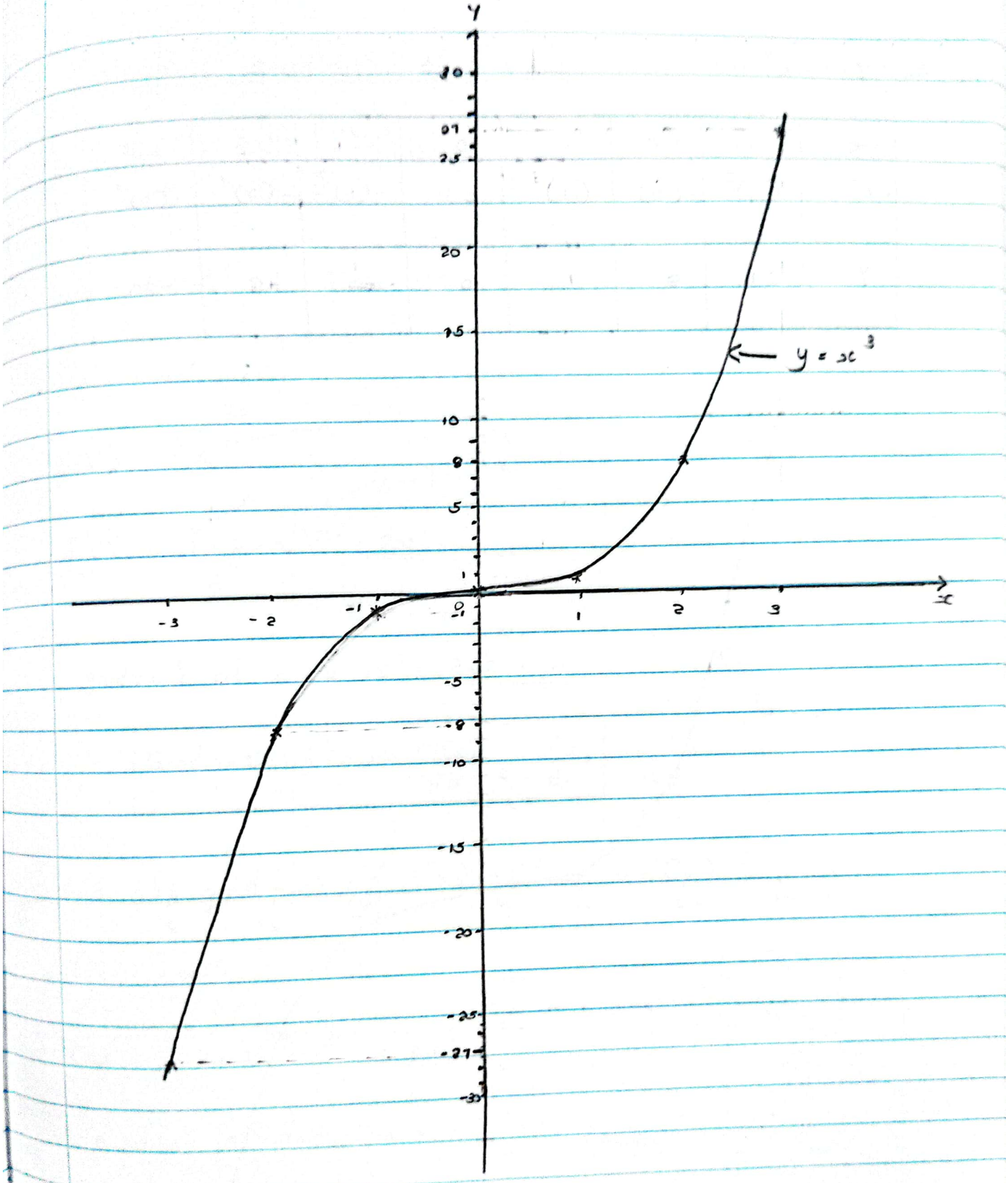
x	-2	-1	0	$+1$	$+2$
$-x^2 + 2x + 3$	$-(-2)^2 + 2(-2) + 3$ $= -4 - 4 + 3$ $= (-5)$	$-(-1)^2 + 2(-1) + 3$ $= -1 - 2 + 3$	$-(0)^2 + 2(0) + 3$	$-1 + 2 + 3$	$-(2)^2 + 2(2) + 3$ $-4 + 4 + 3$
Y	(-5)	0	3	4	3
(x, y)	$(-2, -5)$	$(-1, 0)$	$(0, 3)$	$(1, 4)$	$(2, 3)$

RATNIKA



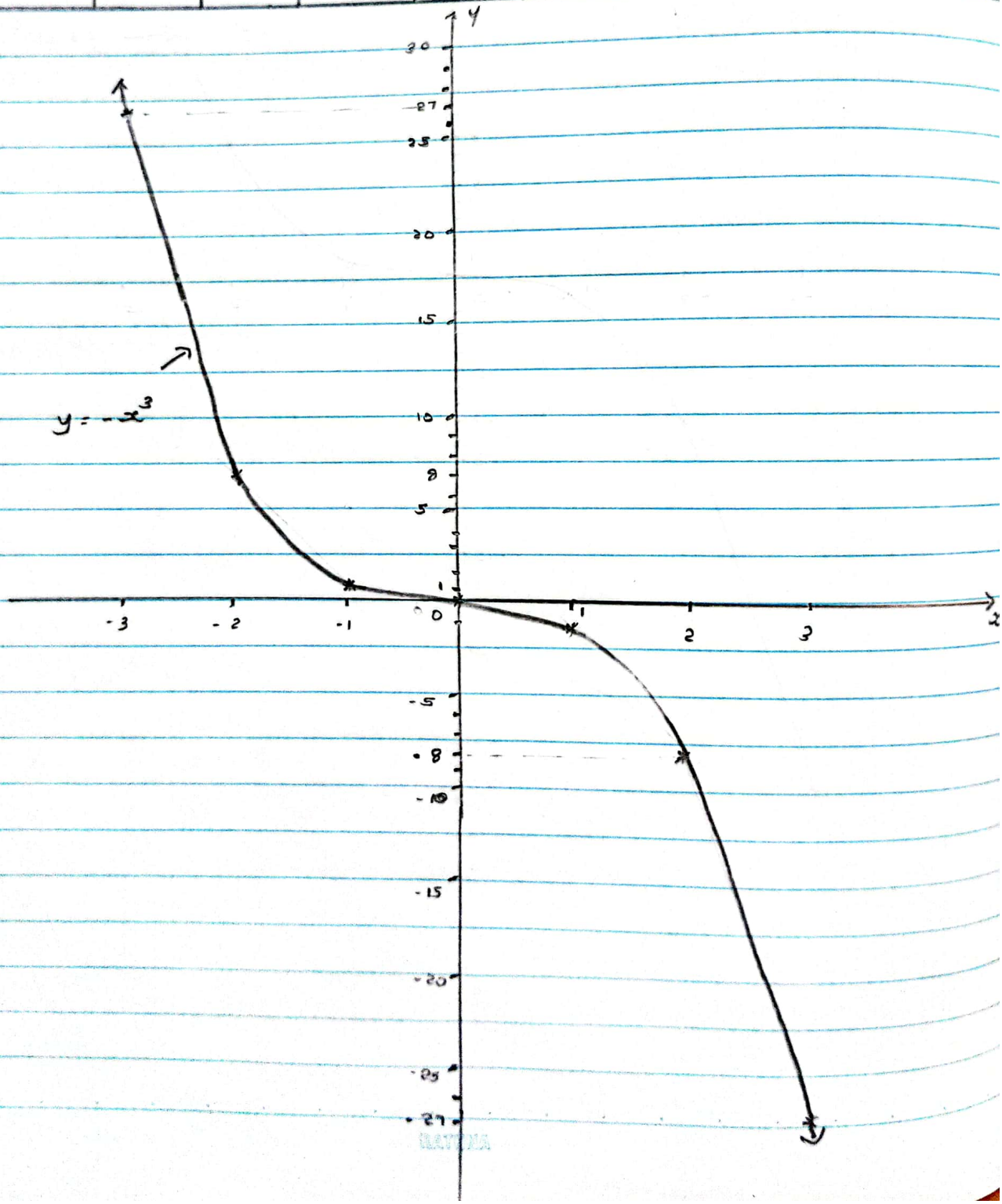
03. a $y = x^3$ -:

x	-3	-2	-1	0	1	2	3
x^3	$(-3)^3$	$(-2)^3$	$(-1)^3$	0	$(1)^3$	$(2)^3$	$(3)^3$
y	-27	-8	-1	0	1	8	27
(x, y)	(-3, -27)	(-2, -8)	(-1, -1)	0	(1, 1)	(2, 8)	(3, 27)


y
30
27
25
20
15
10
8
5
1
-1
-5
-8
-10
-15
-20
-25
-27
-30
-3
-2
-1
0
1
2
3
x
y = x³

04. $y = -x^3$ -:

x	-3	-2	-1	0	1	2	3
$-x^3$	$-(-3)^3$	$-(-2)^3$	$-(-1)^3$	0	$-(1)^3$	$-(2)^3$	$-(3)^3$
y	27	8	1	0	-1	-8	-27



Date 14 | 03 | 2023

Calculate gradient of a graph

Function $f(x)/y$	$f'(x)$ Gradient ← $\frac{d f(x)}{dx}$ $\frac{dy}{dx}$
ax^n	anx^{n-1}

Find gradient of $f(x) = x^2$ at $x = 2$:

- $f(x) = x^2$

$f(x) = 1 x^2$ (a = 1, n = 2)

$f'(x) = 1 \times 2 \times x^{(2-1)} = \underline{2x}$ (a = 1, n = 2, (n-1) = (2-1))

gradient at $x = 2$ is $= 2 \times 2 = \underline{4}$

Gradient $= f'(x) = \frac{df(x)}{dx} = \frac{dy}{dx}$

Find the gradient of $f(x) = x^3$ at $x = (-2)$

- $f(x) = x^3$

- $f'(x) = 1 \times 3x^{(3-1)} = \underline{3x^2}$

gradient at $x = (-2)$ is $= 3 \times (-2)^2 = \underline{12}$

Find the gradient of the graph $y = x^2 - 2x + 1$ at $x = 2$;

* $f(x) = x^2 - 2x + 1$

$f'(x) = 2 \times 1 \times x + (-2) \times 1 \times x^0 + 1 \times 0$ *

$f'(x) = 2x - 2$

note :-
LMS
2

$\therefore$ gradient at $x = 2$ is $2 \times 2 - 2 = \underline{\underline{2}}$

Find the gradient of $f(x) = x^3 + 1$ at point $x = 3$;

* $f(x) = x^3 + 1$

$f'(x) = 3x^2$

$\therefore$ gradient at $x = 3$ is $3 \times (3)^2 = \underline{\underline{27}}$

Calculate the area below the graph

Function	Area below the graph
x^n	$\frac{x^{n+1}}{n+1} + C$

Find the area below $f(x) = x^2$ between $x = 1$ to $x = 2$;

$f(x) = x^2$

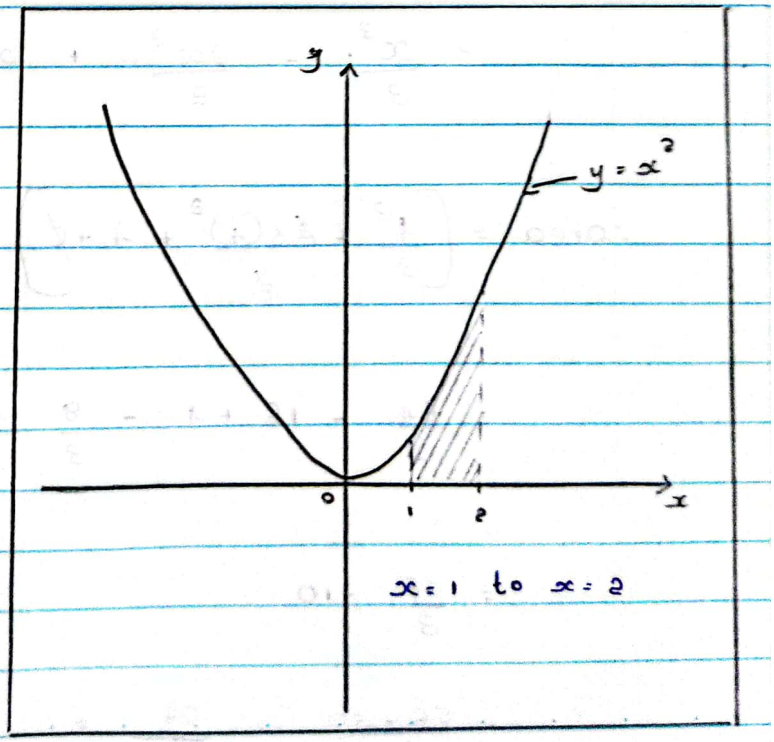
Area $= \frac{x^{2+1}}{2+1} + C$

$= \frac{x^3}{3} + C$

$\therefore$ Area $= \left(\frac{2^3}{3} + C\right) - \left(\frac{1^3}{3} + C\right)$

$= \frac{8}{3} + C - \frac{1}{3} - C$

$= \frac{7}{3}$



Find the area below $f(x) = x^3$ from $x = 1$ to $x = 2$;

* $f(x) = x^3$

$$\text{area} = \frac{x^{3+1}}{3+1} + C$$

$$= \frac{x^4}{4} + C$$

$$\text{area} = \left(\frac{2^4}{4} + C\right) - \left(\frac{1^4}{4} + C\right)$$

$$= \frac{16}{4} + C - \frac{1}{4} - C$$

$$= \frac{15}{4}$$

Find the area below the graph $y = x^2 - 2x + 1$ between $x = 2$ and $x = 4$;

* $y = x^2 - 2x + 1$

$$\text{area} = \frac{x^{2+1}}{2+1} - \frac{2x^{1+1}}{1+1} + \frac{1x^{0+1}}{0+1} + C$$

$$= \frac{x^3}{3} - \frac{2x^2}{2} + x + C$$

$$\therefore \text{area} = \left[\frac{4^3}{3} - \frac{2 \times (4)^2}{2} + 4 + C\right] - \left[\frac{2^3}{3} - \frac{2 \times (2)^2}{2} + 2 + C\right]$$

$$= \frac{64}{3} - 16 + 4 - \frac{8}{3} + 4 - 2$$

$$= \frac{56}{3} - 10$$

$$= \frac{56 - 30}{3} = \frac{26}{3}$$

RATHNA

CamScanner

Find the area below $y = x^2 - x + 1$ from $x = 2$ to $x = 3$;

$y = x^2 - x + 1$

$$\text{area} = \left[\frac{x^{2+1}}{2+1} - \frac{x^{1+1}}{1+1} + \frac{x^{0+1}}{1} + C \right]$$

$$= \frac{x^3}{3} - \frac{x^2}{2} + x + C$$

$$\therefore \text{area} = \left(\frac{3^3}{3} - \frac{(3)^2}{2} + 3 + C \right) - \left(\frac{2^3}{3} - \frac{(2)^2}{2} + 2 + C \right)$$

$$= \frac{27}{3} - \frac{9}{2} + 3 - \frac{8}{3} + \cancel{2} - \cancel{2}$$

$$= \frac{19}{3} + \frac{9}{3} - \frac{9}{2}$$

$$= \frac{28}{3} - \frac{9}{2} = \frac{56}{6} - \frac{27}{6} = \underline{\frac{29}{6}}$$

Limit of a function

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4$$

* $x \neq 2$
* $x \to 2^-$ eg. $x = 1.999\ldots\ldots9$
* $x \to 2^+$ eg. $x = 2.000\ldots\ldots1$

$\left.\begin{matrix}\\ \\ \\ \end{matrix}\right\}$ $\frac{x^2-4}{x-2} \longrightarrow 4$

Find $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$;

$$\frac{4 - 4}{2 - 2} = \frac{0}{0}$$

* $\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2} \frac{x^2 - 2^2}{x - 2}$

$$= \lim_{x \to 2} \frac{(x-2)(x+2)}{(x-2)}$$

$$= \lim_{x \to 2} (x + 2)$$

$$= 4$$

01. $\lim_{x \to 3/2} \frac{4x^2 - 9}{2x - 3} = \lim_{x \to 3/2} \frac{(2x-3)(2x+3)}{(2x-3)}$

$$= \lim_{x \to 3/2} (2x + 3)$$

$$= 6$$

02. $\lim_{x \to 5} \frac{x^2 - 9x + 20}{x^2 - 3x - 10} = \lim_{x \to 5} \frac{x^2 - 5x - 4x + 20}{x^2 - 5x + 2x - 10}$

$= \lim_{x \to 5} \frac{x(x-5) - 4(x-5)}{x(x-5) + 2(x-5)}$

$= \lim_{x \to 5} \frac{(x-4)}{(x+2)}$

$= 1/7$

03. $\lim_{x \to (-5)} \left(\frac{x^2 - 25}{x+5} \right) = \lim_{x \to (-5)} \frac{(x-5)(x+5)}{(x+5)}$

$= \lim_{x \to (-5)} (x-5)$

$= -10$

04. $\lim_{x \to (-1)} \frac{x^4 - x^2}{x+1} = \lim_{x \to (-1)} \frac{x^2(x^2-1)}{(x+1)}$

$= \lim_{x \to (-1)} \frac{x^2(x-1)(x+1)}{(x+1)}$

$= (-2)$

05. $\lim_{x \to (-1)} \frac{x^2 - x - 2}{x^2 + 4x + 3} = \lim_{x \to (-1)} \frac{x^2 - 2x + x - 2}{x^2 + 4x + 3}$

$= \lim_{x \to (-1)} \frac{x(x-2) + 1(x-2)}{(x+1)(x+3)}$

$$= \lim_{x \to (-1)} \frac{(x-2)}{(x+3)}$$

$$= -3/2$$

06. $\lim_{x \to 4} \frac{2 - \sqrt{x}}{4 - x} = \lim_{x \to 4} \frac{(2 - \sqrt{x})(2 + \sqrt{x})}{(4 - x)(2 + \sqrt{x})}$

$$= \lim_{x \to 4} \frac{2^2 - x}{(4 - x)(2 + \sqrt{x})}$$

$$= \lim_{x \to 4} \frac{2 - \sqrt{x}}{2^2 - (\sqrt{x})^2}$$

$$= \lim_{x \to 4} \frac{2 - \sqrt{x}}{(2^2 - \sqrt{x})(2 + \sqrt{x})}$$

$$= 1/4$$

Differentiation

consider the function $f(x) = x^2$. Find $\frac{df(x)}{dx}$ using first principles.

$f(x+h) = (x+h)^2$

$$\frac{df(x)}{dx} = \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} h + 2x$$

$$= 2x$$

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consider the function $f(x) = x^2 - 2x + 1$ find $\frac{df(x)}{dx}$ using first principles.

$$\frac{df(x)}{dx} = \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{(x+h)^2 - 2(x+h) + 1 - (x^2 - 2x + 1)}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{x^2 + 2xh + h^2 - 2x - 2h + 1 - x^2 + 2x - 1}{h} \right)$$

$$= \lim_{h \to 0} \left(\frac{h^2 + 2xh - 2h}{h} \right)$$

$$= \lim_{h \to 0} \left(h + 2x - 2 \right)$$

$$= 2x - 2$$

Revision Tutorial

01. 2, 6, 10, . . . Find,

a) T_{15} ;

$* T_{15} = 2 + (15-1)4$

$= 58$

b) S_{10} ;

$* S_{10} = \frac{1}{2} \times 10 [2 \times 2 + (10-1)4]$

$= 5(40) = 200$

02. 3, 6, 12, . . .

a) T_{10} ;

$* T_{10} = 3 \times 2^{(10-1)}$

$= 3 \times 512$

$= 1536$

b) S_{10} ;

$* S_{10} = \frac{3(1-2^{10})}{(1-2)}$

$= 3069$

03. if $f(x) = 2 \times f(x-1) + 1$ and $f(1) = 12$ then calculate $f(5)$.

$* f(5) = 2f(4) + 1$

$f(4) = 2f(3) + 1$

$f(3) = 2f(2) + 1$

$f(2) = 2f(1) + 1$

$= 2 \times 12 + 1$

$= 25$

$f(3) = 51$

$f(4) = 103$

$\therefore f(5) = 207$

04. $A = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}_{2\times 3}$, $B = \begin{pmatrix} 2 & 5 & 1 \\ 1 & 1 & 0 \end{pmatrix}_{2\times 3}$

① $A + 2B$ -:

* $A + 2B = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}_{2\times 3} + 2\begin{pmatrix} 2 & 5 & 1 \\ 1 & 1 & 0 \end{pmatrix}_{2\times 3}$

$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}_{2\times 3} + \begin{pmatrix} 4 & 10 & 2 \\ 2 & 2 & 0 \end{pmatrix}_{2\times 3}$

$= \begin{pmatrix} 5 & 12 & 5 \\ 5 & 3 & 1 \end{pmatrix}_{2\times 3}$

② AB^{T} -:

* $AB^{T} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}_{2\times 3} \times \begin{pmatrix} 2 & 1 \\ 5 & 1 \\ 1 & 0 \end{pmatrix}_{3\times 2}$

$= \begin{pmatrix} 15 & 3 \\ 12 & 4 \end{pmatrix}_{2\times 2}$

05. Solve simultaneous equations

$x + y = 7$, $2x - y = -1$

if above is written as $A.x = B$ where A, x and B are matrices.

$x + y = 7$

$2x - y = -1$

① $A = \begin{pmatrix} 1 & 1 \\ 2 & (-1) \end{pmatrix}_{2\times 2}$

(2) $A^{-1} = \frac{1}{|A|} \begin{pmatrix} (+1) & (-1) \\ (-3) & 1 \end{pmatrix}$ | $|A| = 1(1) - (3 \times -1)$ $= (+4)$

$= \frac{1}{(4)} \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1/4 & 1/4 \\ 3/4 & (-1/4) \end{pmatrix}_{2 \times 2}$

(3) (a) $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$

$A^{-1} A \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \begin{pmatrix} 7 \\ -1 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1/4 & 1/4 \\ 3/4 & (-1/4) \end{pmatrix} \begin{pmatrix} 7 \\ -1 \end{pmatrix}$

$= \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

$\therefore \quad x = 2 \quad , \quad y = 5$

(b) Convert 29.6875 to

0.6875×2
1.3750

a) Binary :-

$2 \mid 29$	0.6875×2
$2 \mid 14 - 1$	$1 \leftarrow 1.3750$
$2 \mid 7 - 0$	0.3750×2
$2 \mid 3 - 1$	$0 \leftarrow 0.75 \times 2$
$1 - 1$	$1 \leftarrow 1.5$
	0.5×2
	$1 \leftarrow 1.0$

$\therefore 11101.1011_2$

b) octal :-

* 110101.1011 ₂

110 101 . 101 100

2^2 2^1 2^0 2^2 2^1 2^0 2^2 2^1 2^0

6 5 5 4

$= 65.54_8$

c) hexadecimal :-

* 110101.1011

0011 0101 . 1011

2^0 2^1 2^2 2^0 2^3 2^1 2^0

3 5 11

$\therefore 35.B_{16}$

07. consider the function $f(x) = -x^3 + x$

a. find the gradient at $x = 4$ of the graph

* $f'(x) = (-1) \times 3 \times x^2 + 1 \times 1 \times x^0$

$= -3x^2 + 1$

gradient $= -3 \times 16 + 1$

$= -47$

8. find the area below the graph from $x = 2$ to $x = 3$;

area $= -\frac{x^4}{4} + \frac{x^2}{2} + C$

area $= \left(-\frac{3^4}{4} + \frac{3^2}{2} + C\right) - \left(-\frac{2^4}{4} + \frac{2^2}{2} + C\right)$

$= -\frac{81}{4} + \frac{9}{2} + C + \frac{16}{4} - \frac{4}{2} + C$

$= -\frac{63}{4} + 2$

$= -\frac{55}{4}$

$-55/4$ ← L.M.S

9. Consider the number pattern $4k, 2k, k, k/2, \ldots$ where k is a constant. write the 13th value of the above number pattern using k.

$T_{13} = 4k \times \left(\frac{1}{2}\right)^{12} = 4k \times \frac{1}{4096} = \frac{k}{1024}$

$k/1024$ ← L.M.S

10. LED light decoration has bulbs fixed in circles as given in the diagram. Inner most circle has 5 red bulbs.
Next green 8 bulbs.
Next blue 11 bulbs.
again red 14 bulbls, likwise patten continues.
if there are 75 circles.
how many blue bulbs are there.

$$\times 11 \quad 119 \quad 1190 \quad 81 \quad 11 \quad 81 \quad 891 \quad 63 \quad 693 \quad 24 \times 9 = 216$$

$$* \frac{5}{15} = 11 \times \frac{1}{2}\left(2 \times 11 + (25-1)\frac{9}{10}\right)$$

$$= \frac{11}{2}\left(22 + \frac{216}{240}\right)$$

$$= 11\left(11 + \frac{108}{120}\right)$$

$$= \frac{1209}{110}$$